



THE KING'S SCHOOL

2006
Higher School Certificate
Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-10
- All questions are of equal value

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Attempt Questions 1-10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

- | | | |
|-----|---|---|
| (a) | Evaluate, correct to four significant figures $\frac{1251.5 \times \pi}{1.4^2}$ | 2 |
| (b) | Factorise fully $18 - 32a^2$ | 2 |
| (c) | Find a primitive of $4x + \cos 2x$ | 2 |
| (d) | Find the exact value of 225° in radians in simplest form. | 2 |
| (e) | Solve $ 2x - 1 \geq 3$ | 2 |
| (f) | Solve $\frac{x-5}{3} - \frac{x+1}{4} = -5$ | 2 |

End of Question 1

(a) Solve $\tan \theta = -\frac{1}{\sqrt{3}}$ for $0 \leq \theta \leq 2\pi$ 2

(b) Differentiate with respect to x

(i) $2x \cos x$ 2

(ii) $\frac{x^3}{2x + 1}$ 2

(c) (i) Find $\int \frac{2x^3 + 1}{x^4 + 2x} dx$ 2

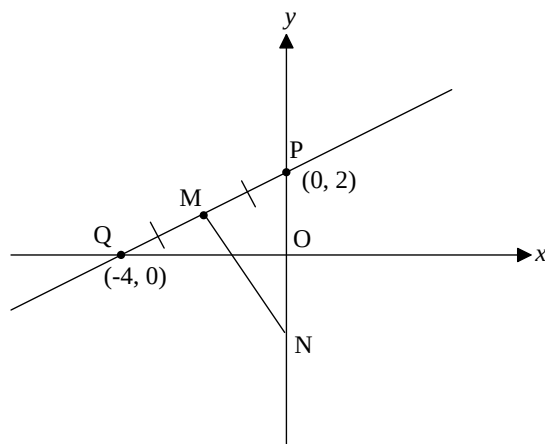
(ii) Evaluate $\int_0^{\frac{\pi}{3}} \sin 3x \, dx$ 2

(d) Solve the simultaneous equations

$$x + 2y = -2 \text{ and } 2x - y = 11$$
2

End of Question 2

(a)



The diagram shows the points P (0, 2) and Q (-4, 0). The point M is the midpoint of PQ. The line MN is perpendicular to PQ and meets the y axis at N.

- | | |
|--|----------|
| (i) Show that the gradient of PQ is $\frac{1}{2}$. | 1 |
| (ii) Find the coordinates of M. | 2 |
| (iii) Find the equation of the line MN. | 2 |
| (iv) Show that N has coordinates (0, -3). | 1 |
| (v) Find the distance NQ. | 1 |
| (vi) The point R lies in the second quadrant, and PNQR is a rhombus.
Find the coordinates of R. | 1 |
| (vii) Find the area of rhombus PNQR. | 1 |

- | | |
|--|----------|
| (b) Find the equation of the tangent to the curve $y = e^{x^2}$ at the point where $x = 1$. | 3 |
|--|----------|

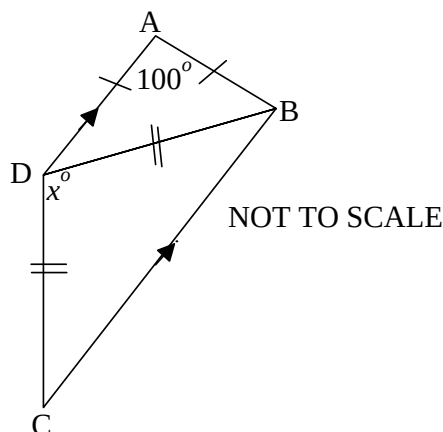
End of Question 3

- (a) Find the value of x giving reasons for your answer.

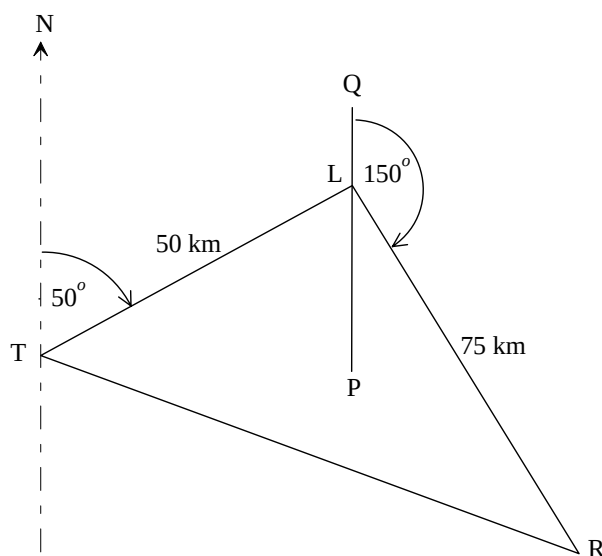
3

$$AB = AD$$

$$BD = DC \text{ and } AD \parallel BC$$



- (b)



A motorist drives 50 km from town T to a landmark L on a bearing of 050° .
He then drives 75 km to a resort R on a bearing of 150° .

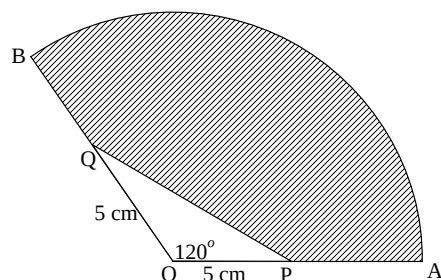
- (i) Find the size of $\angle TLR$. 1
- (ii) Find the distance TR to the nearest kilometre 2

- (c) Solve $\log_2 x + \log_2 (x - 1) = 1$ 3

- (d) Find $\frac{dy}{dx}$ when $y = \ln \sqrt{\frac{4x - 1}{2x + 3}}$ 3

End of Question 4

(a)



O is the centre of the circle containing the arc AB. P is the midpoint of OA and Q is the midpoint of OB.

$$\angle AOB = 120^\circ \quad OQ = OP = 5 \text{ cm}$$

- (i) Find the exact length of the arc AB. 1
- (ii) Find the shaded area PQBA in exact form. 2

(b) Find, justifying your answer, any point(s) of inflexion on the curve

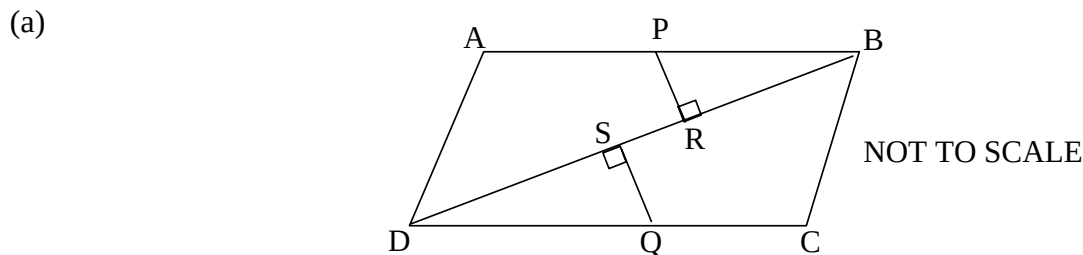
$$y = \frac{1}{3}x^3 - x^2 + 1 \quad \text{3}$$

(c) Water is taken out of a tank at the rate $R = 2t + \frac{200}{t + 4}$ litres/min.

Initially there were 1000 litres of water in the tank.

- (i) At what time will the rate be a minimum? 3
- (ii) How many litres of water will be left in the tank after $24\frac{1}{2}$ minutes. 3

End of Question 5

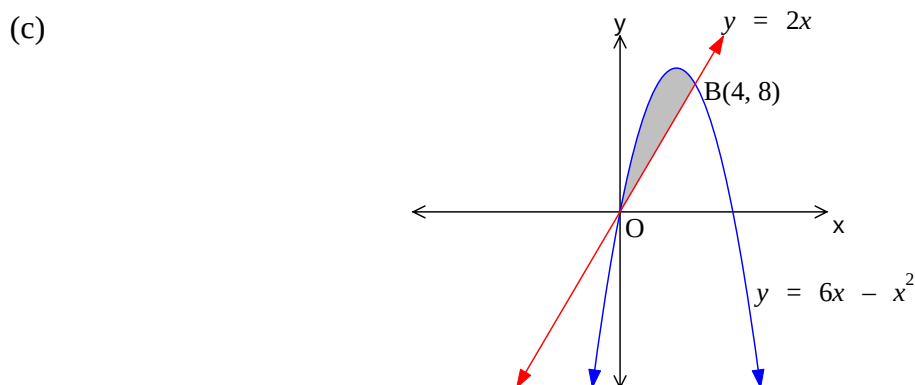


ABCD is a parallelogram and P, Q are the midpoints of the sides AB, DC respectively. The intervals PR and QS are drawn perpendicular to the diagonal BD.

- (i) Prove the triangles BPR and DQS are congruent. 2
- (ii) If $AB = 10$ cm, $PR = 3$ cm and $BD = 14$ cm, find the length of SR. 2

(b) The equation $(x - 2)^2 = -6\left(y - \frac{3}{2}\right)$ represents a parabola.

- (i) Write down the coordinates of the vertex. 1
- (ii) Find the focal length. 1
- (iii) Find the x -intercepts of the parabola. 2
- (iv) Sketch the graph of the parabola showing the focus and directrix. 1



The graphs of $y = 2x$ and $y = 6x - x^2$ intersect at the origin and at point B (4, 8).

Find the shaded area bounded by $y = 6x - x^2$ and $y = 2x$. 3

End of Question 6

- (a) The table shows the values where $y = 5xe^{-x}$.

x	0	1	2	3	4
y	0	1.84	1.35		0.37

- (i) Find y when $x = 3$ correct to two decimal places. **1**
- (ii) Find $\int_0^4 5x e^{-x} dx$ using Simpson's Rule with five function values, correct to one decimal place.. **2**
- (b) The number, N , of bacteria in a culture is growing exponentially according to the formula
- $$N = 120e^{kt}$$
- (i) What is the initial number of bacteria? **1**
- (ii) After eight hours the number of bacteria has tripled. Calculate the value of k to three decimal places. **2**
- (iii) How many bacteria, correct to the nearest ten, will there be after twelve hours? **1**
- (iv) At what rate will the bacteria be increasing after eighteen hours? **2**
- (c) Sketch the curve $y = 1 - \sin 2x$ for $0 \leq x \leq \pi$ **3**

End of Question 7

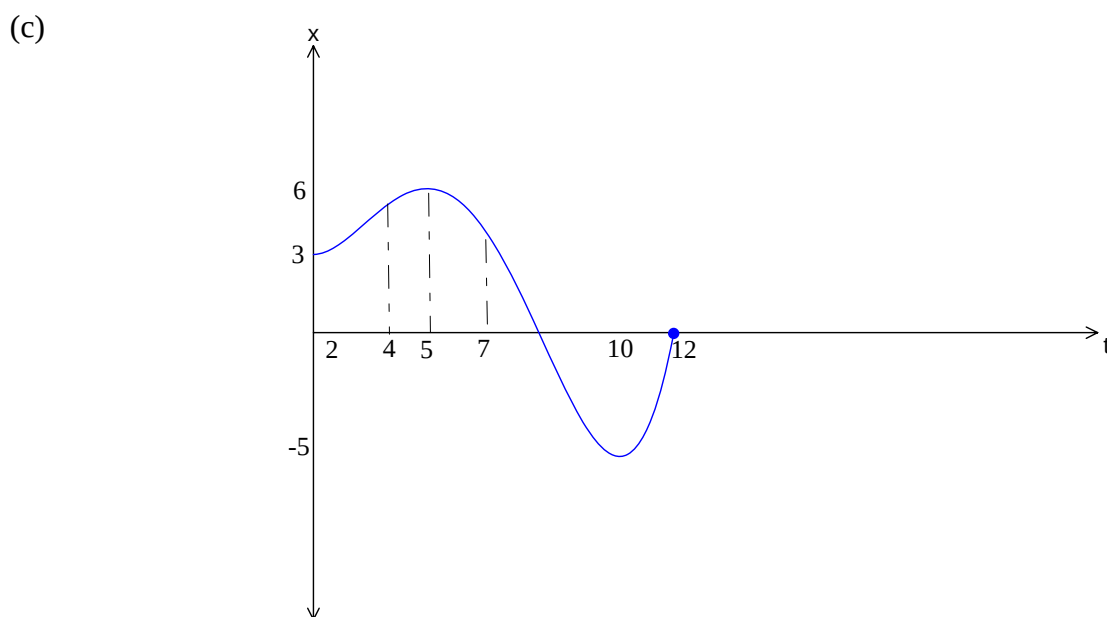
- (a) (i) **Without** using calculus, sketch the curve $y = e^x - 2$ 1
- (ii) On the same sketch, find graphically the number of solutions of the equation

$$e^x - x - 2 = 0$$

(Do not attempt to solve the equation.) 2

- (b) Some children have 500 identical toy blocks. They wish to build a triangular wall so that the top layer has 1 brick, the second layer has 2 bricks, the third layer has 3 bricks and so on. Assuming that the wall remains stable, find

- (i) the maximum number of layers they can complete 3
- (ii) the number of bricks they have left over 1



A particle moves along the x axis. The graph shows the displacement x metres of the particle, from a fixed point (origin) at time t seconds.

- (i) What is the initial velocity of the particle? 1
- (ii) Between what times is the particle moving in a negative direction? 1
- (iii) How many metres did the particle travel? 1
- (iv) Sketch the velocity as a function of time. 2

End of Question 8

- (a) What are the values of a and b if $x - 3$ and $x + 7$ are factors of the quadratic expression

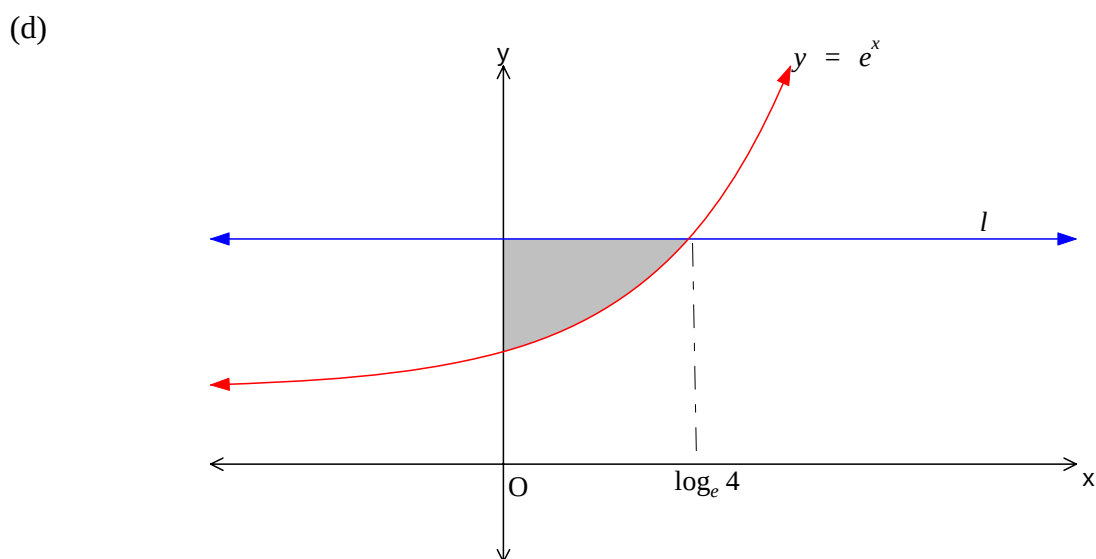
$$ax^2 + 12x + b \quad 2$$

- (b) A man sets up a trust fund for his grandson by investing \$75 at the beginning of each month. The money is invested at 6% p.a. compounded monthly and it matures at the end of the month, 20 years after the first investment.

- (i) After 20 years, what will be the value of the first \$75 invested? 1

- (ii) Using geometric series, calculate the final value of the trust fund at the end of the 20 years. 3

- (c) Solve $2\cos^2 x - 3\cos x - 2 = 0$ for $0 \leq x \leq 2\pi$ 3



In the diagram, the shaded region is bounded by the y axis, the curve $y = e^x$ and a horizontal line l that cuts the curve at a point whose x coordinate is $\log_e 4$.

A solid is formed by rotating the shaded region about the y axis.

Find a definite integral in terms of y whose value is the volume of the solid formed. 3

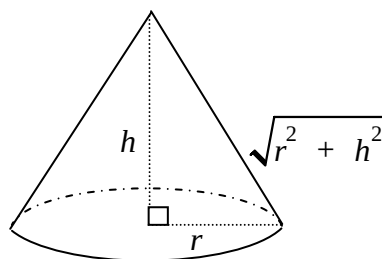
(Do not attempt to evaluate the integral.)

End of Question 9

(a) (i) Prove that $\frac{d}{d\theta} (\tan^3 \theta) = 3 \sec^4 \theta - 3 \sec^2 \theta$ 2

(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \sec^4 \theta d\theta$ 2

(b)



A right circular cone has base radius r and height h .

As r and h vary, its curved surface area $\pi r \sqrt{r^2 + h^2}$ is kept constant. Let $\pi r \sqrt{r^2 + h^2} = K$ where K is a constant.

(i) Show that $V^2 = \frac{1}{9} r^2 (K^2 - \pi^2 r^4)$ where V is the volume of the cone. (Note: $V = \frac{1}{3} \pi r^2 h$) 2

(ii) Let $Q = V^2$. Show that $\frac{dQ}{dr} = 0$ when $r^4 = \frac{K^2}{3\pi^2}$ 2

(iii) Show that the maximum value of Q (and hence the maximum value of V) occurs when $h = \sqrt{2} r$. 4

End of Examination

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 + x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$

Year 12 - 2 Unit Mathematics Aug 2006

Question 1

(a) $E_{np} = 2005.971$ — 1 mark
 $= 2006$ (to 4 sig fig) — 1 mark

(b) $18 - 32a^2 = 2(9 - 16a^2)$ — 1 mark
 $= 2(3 + 4a)(3 - 4a)$ — 1 mark

(c) $2x^2 + \frac{1}{2} \sin 2x (+c)$ 1 off each error

(d) $225^\circ = \frac{\pi}{180} \times 225$ radians — 1 mark
 $= \frac{5\pi}{4}$ radians — 1 mark

(e) $2x - 1 \leq -3$ $2x - 1 \geq 3$
 $2x \leq -2$ $2x \geq 4$ — 1 each
 $x \leq -1$ $x \geq 2$
 $\therefore x \leq -1$ or $x \geq 2$

(f) $\frac{x-5}{3} - \frac{x+1}{4} = -5$
 $4(x-5) - 3(x+1) = -60$ 1 off each error
 $4x - 20 - 3x - 3 = -60$
 $x - 23 = -60$
 $x = -37$

Question 2

(a) $\tan \theta = -\frac{1}{\sqrt{3}}$
 $\therefore$ acute $\theta = \frac{\pi}{6}$ — 1 mark
 $\theta = \pi - \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$
 $= \frac{5\pi}{6}, \frac{11\pi}{6}$ — 1 mark

(b) i, $\frac{d}{dx}(2x \cos x) = \cos x \cdot 2 + 2x \cdot -\sin x$ — 1 mark
 $= 2 \cos x - 2x \sin x$ — 1 mark

$$(ii), \frac{d}{dx} \left(\frac{x^3}{2x+1} \right) = \frac{(2x+1) \cdot 3x^2 - x^3 \cdot 2}{(2x+1)^2}$$

$$= \frac{6x^3 + 3x^2 - 2x^3}{(2x+1)^2}$$

$$= \frac{4x^3 + 3x^2}{(2x+1)^2} \quad -1 \text{ off each error}$$

$$(c) (i), \int \frac{2x^3 + 1}{x^4 + 2x} dx = \frac{1}{2} \int \frac{4x^3 + 2}{x^4 + 2x} dx \quad -1 \text{ mark}$$

$$= \frac{1}{2} \log(x^4 + 2x) + C \quad -1 \text{ mark}$$

$$(ii), \int_0^{\pi} \sin 3x dx = -\frac{1}{3} [\cos 3x]_0^{\pi}$$

$$= -\frac{1}{3} [\cos \pi - \cos 0]$$

$$= -\frac{1}{3} \times -2$$

$$= \frac{2}{3} \quad -1 \text{ off each error}$$

$$(d), \quad x + 2y = -2 \quad \dots \quad (1)$$

$$2x - y = 11 \quad \dots \quad (2)$$

$$(2) \times 2 \quad 4x - 2y = 22 \quad \dots \quad (3)$$

$$(1) + (3) \quad 5x = 20$$

$$x = 4$$

sub in (2)

$$2x - y = 11$$

$$y = -3$$

-1 off each error

$$\begin{cases} x = 4 \\ y = -3 \end{cases}$$

Question 3

$$(a) \quad i, \quad m = \frac{2-0}{0-4}$$
$$= \frac{2}{-4}$$
$$= -\frac{1}{2} \quad -1 \text{ mark}$$

$$(ii), \text{ coords of } M = \left(\frac{0+4}{2}, \frac{2+0}{2} \right)$$
$$= (-2, 1) \quad -1 \text{ mark each coordinate}$$

$$(iii), \quad y-1 = -2(x-2)$$
$$= -2x+4$$
$$y = -2x+3 \quad -1 \text{ off each line}$$

$$(iv), \text{ cuts } y\text{-axis where } x=0$$

$$y = 0-3$$
$$= -3$$
$$\text{coordinates of } N \text{ are } (0, -3) \quad -1 \text{ mark}$$

$$(v) \quad NQ = \sqrt{(-4-0)^2 + (0-3)^2}$$
$$= \sqrt{25}$$
$$= 5 \text{ units} \quad -1 \text{ mark}$$

$$(vi), \quad PN = QR$$
$$= 5 \text{ units}$$
$$\text{Coordinates of } R \text{ are } (-4, 5) \quad -1 \text{ mark}$$

$$(vii), \quad \text{area} = bh$$
$$= 4 \times 5$$
$$= 20 \text{ u}^2 \quad -1 \text{ mark}$$

$$(b), \quad \frac{dy}{dx} = e^{x^2} \cdot 2x$$
$$= 2x \cdot e^{x^2} \quad -1 \text{ mark}$$

where $x=1$, $y=e$

$$\frac{dy}{dx} = 2e \quad - 1 \text{ mark}$$

$$y - e = 2e(x-1)$$
$$= 2ex - 2e$$

$$y = 2ex - e \quad - 1 \text{ mark}$$

Question 4

(a) $\angle ADB = \angle BDC$ (base $\angle$ s, isos $\triangle ADB$)
 $= 40^\circ \quad - 1 \text{ mark}$

$\angle DCB = 40^\circ$ ($\triangle DCB$ is isos) $- 1 \text{ mark}$

$\therefore \angle C = 100^\circ$ ($\angle$ sum of $\triangle DCB$) $- 1 \text{ mark}$

(b) (i) $\angle TLP = 50^\circ$ (alt $\angle$ $NT \parallel QP$)

$\angle PLR = 30^\circ$ ($\angle QLP$ alt $\angle$)

$\therefore \angle TLR = 80^\circ \quad - 1 \text{ mark}$

(ii) $TR^2 = 50^2 + 75^2 - 2 \times 50 \times 75 \times \cos 50^\circ \quad - 1 \text{ mark}$
 $= 6822.6387$

$\therefore TR = 83 \text{ km}$ (nearest km) $- 1 \text{ mark}$

(c) $\log_2 x + \log_2 (x-1) = 1$
 $\therefore x(x-1) = 2 \quad - 1 \text{ mark}$
 $x^2 - x - 2 = 0$

$$(x-2)(x+1) = 0$$

$$x = 2, -1 \quad - 1 \text{ mark}$$

$$\therefore x = 2, \quad x \neq -1 \quad - 1 \text{ mark}$$

(d) $y = \ln \sqrt{\frac{4x-1}{2x+3}}$

$$= \frac{1}{2} \ln \frac{4x-1}{2x+3}$$

$$= \frac{1}{2} \{ \ln(4x-1) - \ln(2x+3) \} \quad - 1 \text{ off each corner}$$

$$\frac{dy}{dx} = \frac{1}{2} \left(\frac{4}{4x-1} - \frac{2}{2x+3} \right)$$

$$= \frac{2}{4x-1} - \frac{1}{2x+3}$$

Question 5

(a) i, arc length = $\frac{2\pi}{3} \times 10$ cm
 $= \frac{20\pi}{3}$ cm — 1 mark

ii, shaded area = $\frac{1}{2} \times 10^2 \times \frac{2\pi}{3} - \frac{1}{2} \times 5 \times 5 \times \sin \frac{2\pi}{3}$
 — 1 mark

$$= \left(\frac{100\pi}{3} - \frac{25\sqrt{3}}{4} \right) \text{cm}^2 \quad \text{— 1 mark}$$

(b) i, $y = \frac{1}{3}x^3 - x^2 + 1$

$$\frac{dy}{dx} = x^2 - 2x$$

$$\frac{d^2y}{dx^2} = 2x - 2 \quad \text{— 1 mark}$$

where $\frac{d^2y}{dx^2} = 0$, $x = 1$, $y = \frac{1}{3}$. — 1 mark

$$x = \frac{1}{2}, \quad \frac{d^2y}{dx^2} = 1 - 2 < 0$$

$$x = 1\frac{1}{2}, \quad \frac{d^2y}{dx^2} = 3 - 2 > 0 \quad \text{— 1 mark}$$

$\therefore$ pt of inflexion at $(1, \frac{1}{3})$

(c) ii) $R = 2t + 200(t+4)^{-1}$

$$\frac{dR}{dt} = 2 - \frac{200}{(t+4)^2} \quad \text{— 1 off each error}$$

where $\frac{dR}{dt} = 0$, $2 = \frac{200}{(t+4)^2}$

$$(t+4)^2 = 100$$

$$t+4 = \pm 10$$

$$t = 6, \quad t \neq -14.$$

$$\frac{d^2r}{dt^2} = \frac{400}{(t+4)^3}$$

$$> 0, \quad t > 0$$

$\therefore$ min at $t = 6$

$$\text{ii, Vol emptied} = \int (2t + \frac{200}{t+4}) dt$$

- 1 off each error

$$= t^2 + 200 \log(t+4) + C.$$

$$\text{where } t=0, \quad v=0$$

$$\therefore V = t^2 + 200 \log(t+4) - 200 \log 4$$

$$\text{where } t = 24\frac{1}{2}, \quad V = (24\frac{1}{2})^2 + 200 \log 28\frac{1}{2} - 200 \log 4$$

$$= 992.97$$

$$\therefore \text{water left} = 7 \text{ litres (nearest litre)}$$

Question 6

$$(a) \text{ ii, } \triangle BPR, \triangle DQS$$

$$BP = DQ \text{ (half of equal sides of } \triangle PQR)$$

$$\angle PBR = \angle SDQ \text{ (alt } \angle s, AB \parallel DC)$$

$$\angle PRB = \angle DSQ \text{ (both } \angle s)$$

$$\therefore \triangle BPR \equiv \triangle DQS \text{ [AAS]}$$

- 1 off each error

$$\text{ii, By Pythagoras } BR^2 = 5^2 - 3^2$$

$$= 16$$

$$BR = 4 \text{ cm}$$

$$\text{similarly } DS = 4 \text{ cm}$$

$$\therefore SR = 6 \text{ cm}$$

- 1 off each error

(b) i, vertex $(2, 1\frac{1}{2})$ - 1 mark

ii, focal length $= 1\frac{1}{2}$ units - 1 mark

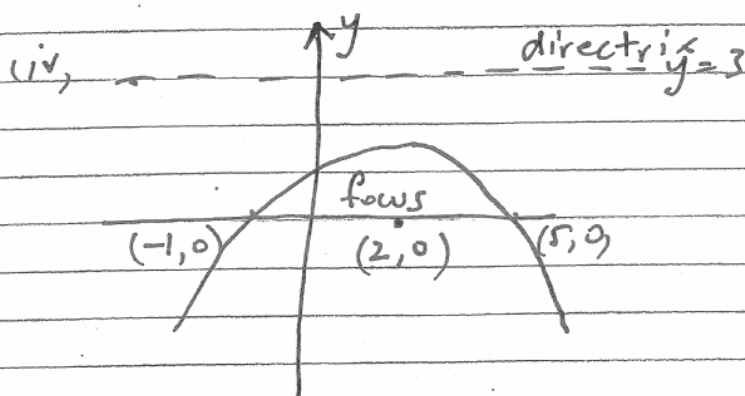
iii, x-intercept where $y=0$

$$(x-2)^2 = 9 \quad - 1 \text{ mark}$$

$$x-2 = \pm 3$$

$$x = 5, -1$$

$\therefore$ intercepts at $(5,0)$ and $(-1,0)$ - 1 mark



- 1 mark

(c) Reg'd area $= \int_0^4 (6x - x^2 - 2x) dx$

$$= \int_0^4 (4x - x^2) dx$$

$$= \left[2x^2 - \frac{x^3}{3} \right]_0^4$$

- 1 off each error

$$= \left[32 - \frac{64}{3} \right] - [0]$$

$$= 10\frac{2}{3} \text{ m}^2$$

Question 7

(a) i) 0.75 - 1 mark

ii) $\int_0^4 5xe^{-x} dx = \frac{1}{3} \{ 0 + 4 \times 1.84 + 2 \times 1.35 + 4 \times 0.75 + 0.37 \}$

- 1 mark

$$= 4.5 \text{ (to 1 dec pl)} - 1 \text{ mark}$$

(b) i, where $t=0$, $N=120$. — 1 mark

ii, $360 = 120 e^{sk}$
 $3 = e^{sk}$

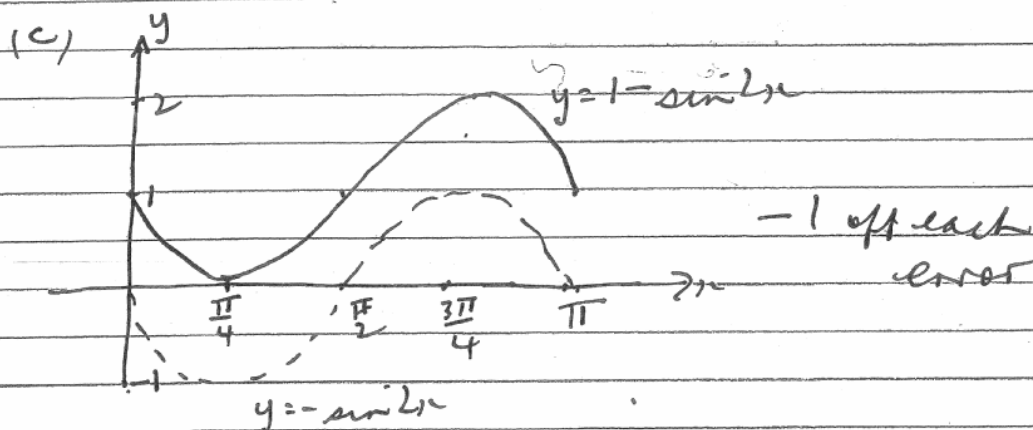
$\log_e 3 = sk$ — 1 mark

$k = 0.137$ (to 3 dec pl) — 1 mark

iii, where $t=12$, $N = 120 e^{0.137 \times 12}$ OR $N = 120 e^{\frac{1}{2} \ln 3 \times 12}$
 $= 620$ (nearest 10) — 1 mark

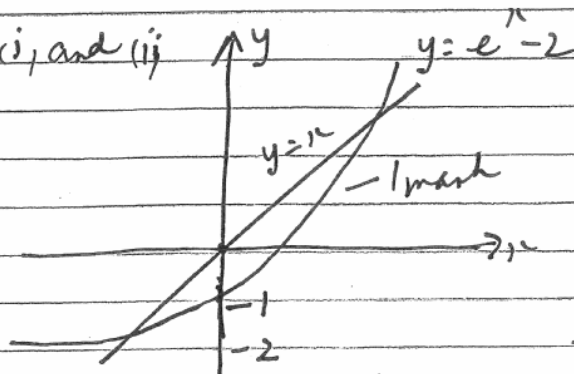
iv, $\frac{dN}{dt} = 120 e^{0.137t} \times 0.137$ OR $\frac{dN}{dt} = 120 e^{\frac{1}{2} \ln 3 t} \times \frac{1}{2} \ln 3$ — 1 mark

where $t=18$, $\frac{dN}{dt} = 194$ $t=18$ $\frac{dN}{dt} = 195$ (nearest whole number)
 — 1 mark



Question 8

(a) i, and ii



$y = x$
 $y = e^x - 2$ — 1 mark

$\therefore e^x - x - 2 = 0$ — 1 mark

$\therefore 2$ solutions

(b) i) For $S = 500$

$$\frac{n}{2} [2 + (n-1)1] = 500$$

$$\frac{n}{2} (n+1) = 500$$

$$n^2 + n - 1000 = 0$$

$$n = \frac{-1 \pm \sqrt{1 - 4 \times 1 \times -1000}}{2}$$

$$= \frac{-1 \pm \sqrt{4001}}{2}$$

$$= 31.13, -32.13$$

$\therefore$ but n is a positive integer

$$\therefore n = 31$$

i.e. 31 layers.

- 1 off each
error

ii, $S = \frac{31}{2} (1+31)$
 $= 496$

- 1 mark

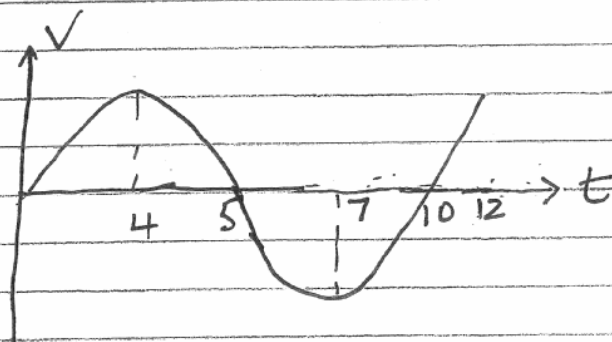
$\therefore$ 4 bricks left over

(c) i) initial velocity = 0 m/s - 1 mark

ii, between 5th and 10th seconds - 1 mark

iii, particle moves 19m - 1 mark

(iv)



- 1 mark
for basic
shape

- 1 mark for
intercepts at
5 and 10

Question 9

$$(a), ax^2 + 12x + b = a(x-2)(x+7) \\ = ax^2 + 4ax - 21a \quad -1 \text{ mark}$$

$$\therefore 4a = 12$$

$$a = 3$$

$$b = -21a$$

$$= -63$$

$$\begin{cases} a = 3 \\ b = -63 \end{cases} \quad -1 \text{ mark}$$

$$(b) \text{ i), } A_1 = 75 \times 1.005^{240} \quad -1 \text{ mark} \\ = \$248.27$$

$$\text{ii), } A_5 = 75 \left[\frac{1.005(1.005^{240} - 1)}{1.005 - 1} \right] \quad -1 \text{ off each error}$$

$$= \$34826.63$$

$$(c), 2\cos^2 x - 3\cos x - 2 = 0$$

$$(2\cos x + 1)(\cos x - 2) = 0 \quad -1 \text{ mark}$$

$$\cos x = -\frac{1}{2}, \cos x \neq 2 \quad -1 \text{ mark}$$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad -1 \text{ mark}$$

$$(d) \text{ where } x = \log_x 4, y = e^{\log 4} \\ = 4 \quad -1 \text{ mark off}$$

$$x = 0, y = 1$$

each error

$$\therefore V = \pi \int_1^4 x^2 dy$$

$$y = e^x$$

$$\therefore x = \log_e y$$

$$V = \pi \int_1^4 (\log_e y)^2 dy$$

Question 10

$$\begin{aligned} \text{(a, i), } \frac{d}{d\theta} (\tan^3 \theta) &= 3 \tan^2 \theta \sec^2 \theta \quad -1 \text{ mark} \\ &= 3 \sec^2 \theta (\sec^2 \theta - 1) \\ &= 3 \sec^4 \theta - 3 \sec^2 \theta \quad -1 \text{ mark} \end{aligned}$$

$$\begin{aligned} \text{ii, } \int_0^{\frac{\pi}{4}} \sec^4 \theta d\theta &= \frac{1}{3} \int_0^{\frac{\pi}{4}} (3 \sec^4 \theta - 3 \sec^2 \theta) d\theta \\ &\quad + \frac{1}{4} \int_0^{\frac{\pi}{4}} \sec^2 \theta d\theta \quad -1 \text{ mark} \\ &= \frac{1}{3} [\tan^3 \theta]_0^{\frac{\pi}{4}} + [\tan \theta]_0^{\frac{\pi}{4}} \\ &= \frac{1}{3} + 1 \quad -1 \text{ mark} \\ &= \frac{4}{3} \end{aligned}$$

$$\begin{aligned} \text{(b) iv } \pi^2 r^2 (r^2 + h^2) &= K^2 \\ \pi^2 r^4 + \pi^2 r^2 h^2 &= K^2 \end{aligned}$$

$$h^2 = \frac{K^2 - \pi^2 r^4}{\pi^2 r^2} \quad -1 \text{ mark}$$

$$\begin{aligned} V &= \frac{1}{3} \pi r^2 h \\ \therefore V &= \frac{1}{9} \pi r^4 \times \frac{(K^2 - \pi^2 r^4)}{\pi^2 r^2} \end{aligned}$$

$$= \frac{1}{9} r^2 (K^2 - \pi^2 r^4) \quad -1 \text{ mark}$$

$$\text{iii, } Q = \frac{1}{9} K^2 r^2 - \frac{1}{9} \pi^2 r^6$$

$$\frac{dQ}{dr} = \frac{2}{9} K^2 r - \frac{2}{3} \pi^2 r^5$$

$$= \frac{2}{9} r (K^2 - 3\pi^2 r^4) \quad -1 \text{ mark}$$

$$\text{where } \frac{dQ}{dr} = 0, \quad K^2 - 3\pi^2 r^4 = 0$$

$$K^2 = 3\pi^2 r^4 \quad -1 \text{ mark}$$

$$r^4 = \frac{K^2}{3\pi^2}, \quad r \neq 0$$

$$\text{iii, } \frac{d^2 \alpha}{dr^2} = \frac{2}{9} k^2 - \frac{10\pi^2}{3} r^{-4}$$

$$\text{where } r^4 = \frac{k^2}{3\pi^2} \quad \frac{d^2 \alpha}{dr^2} = \frac{2}{9} k^2 - \frac{10\pi^2}{3} \times \frac{k^2}{3\pi^2}$$

$$= \frac{2k^2}{9} - \frac{10k^2}{9}$$

$$\therefore \text{max at } r^4 = \frac{k^2}{3\pi^2} \quad \text{--- 1 off each error}$$

$$\text{but } k^2 = \pi^2 r^2 (r^2 + h^2)$$

$$\therefore r^4 = \frac{\pi^2 r^2 (r^2 + h^2)}{3\pi^2}$$

$$3r^4 = r^4 + r^2 h^2$$

$$2r^4 = r^2 h^2$$

$$h^2 = 2r^2$$

$$\therefore h = \sqrt{2}r, \quad \therefore r \text{ and } h > 0$$



THE KING'S SCHOOL

2006 Higher School Certificate Trial Examination

Mathematics

Question	Algebra and Number	Geometry	Functions	Trigonometry	Differential Calculus	Integral Calculus	Total
1	(a), (b), (e), (f) /8			(d) /2		(c) /2	12
2	(d) /2			(a) /2	(b) /4	(c) /4	12
3			(a) /9		(b) /3		12
4	(c) /3	(a) (3)		(b) /3	(d) /3		12
5				(a) /3	(b) /3	(c) /6	12
6		(a) /4	(b) /5			(c) /3	12
7				(c) /3	(b) /6	(a) /3	12
8	(b) /4		(a) /3		(c) /5		12
9	(b) /4		(a) /2	(c) /3		(d) /3	12
10					(b) /8	(a) /4	12
Marks	/21	/7	/19	/16	/33	/24	120